



Cryptographic reduction bound and operational audit boundary

$$\text{Adv}^*(A) \leq \text{Adv_PH-MA-ABE}(B1) + \text{Adv_COM}(B2) + \text{Adv_PAD}(B3) + \text{Adv_SIG}(B4) \\ + \text{Adv_CRED}(B5a) + \text{Adv_ZK}(B5b) + \text{negl}(\lambda)$$

B1: policy/attribute hiding

B2: commitment consistency

B3: request-type padding

B4: audit-chain unforgeability

B5a: key-credential forgery

B5b: proof soundness

Operational audit, outside the advantage bound: $\text{RiskGap}(R) \leq \tau_R$